

Method

Base-38 encoding is achieved by employing a simplified strategy where every 3 bytes (24 bits) of binary source data are encoded to 5 characters of the Base-38 alphabet.

Data from the Packed Binary Data Structure are encoded starting with the first byte of the structure. Three-byte chunks are formed into a 24-bit unsigned integer for encoding as follows:

$$\text{UINT}_{24} = (\text{BYTE}_{N+2} \ll 16) \mid (\text{BYTE}_{N+1} \ll 8) \mid (\text{BYTE}_N \ll 0)$$

The 24-bit value is subsequently converted to Base-38 radix using the alphabet above to produce a 5-character substring, with the least-significant character appearing first (little-endian).

If a single byte of binary source data remains, it SHALL be converted to Base-38 radix using the alphabet above to produce a 2-character substring, with the least-significant character appearing first.

If two bytes of binary source data remains, they SHALL be formed into a 16-bit unsigned integer for encoding as follows:

$$\text{UINT}_{16} = (\text{BYTE}_{N+1} \ll 8) \mid (\text{BYTE}_N \ll 0)$$

This 16-bit value is subsequently converted to Base-38 radix using the alphabet above to produce a 4-character substring, with the least-significant character appearing first.

The final encoded string is a result of concatenation of all substrings, with the first-encoded substring appearing at the beginning of the concatenated string.

5.1.3.2. QR Code Format

The format selection, which includes the QR code Version and ECC levels as well as size and color, MAY be tailored to the requirements of the manufacturer and their respective product, provided it meets the following requirements:

QR code Version and Encoding

The QR code generated, as defined in [ISO/IEC 18004:2015](#), SHALL be of Version 1 or higher, using alphanumeric encoding. The size of the payload implies a minimum Version, though a higher Version may be needed to allow a higher ECC level. For example, a minimum payload of 22 alphanumeric characters (19 base-38-encoded characters from the packed binary structure plus 3 prefix characters) can be fit into a Version 1 with ECC=L, but for ECC=M, Q or H, the same payload requires a Version 2 QR code. This allows the Manufacturer to balance between ECC, pixel size and overall size.

Example QR Code Sizes and Payloads